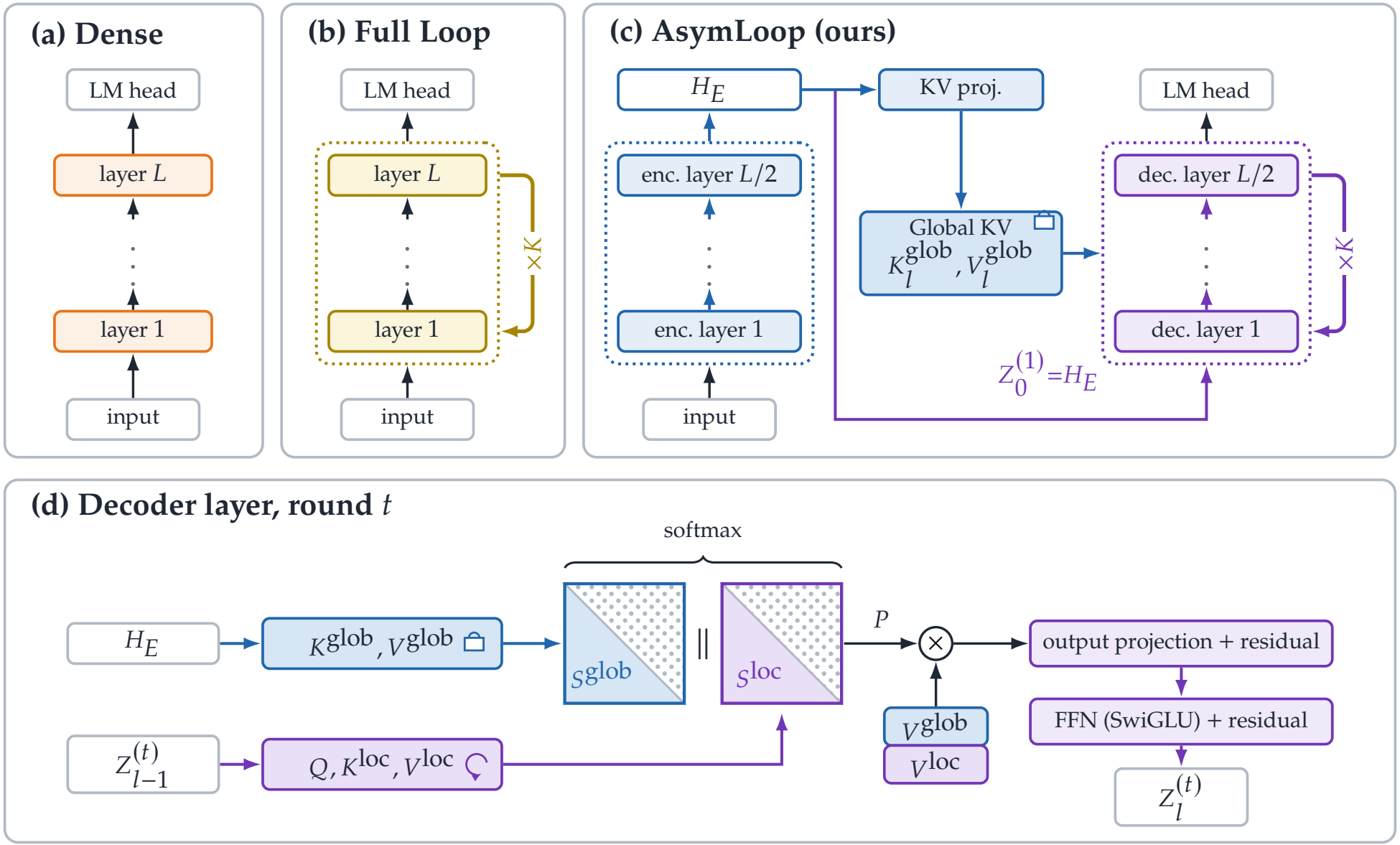

(a) Dense
LM head
layer L
layer 1
input
(b) Full Loop
LM head
layer L
layer 1
input
$\times K$
(c) AsymLoop (ours)
H_E
enc. layer $L/2$
enc. layer 1
input
KV proj.
Global KV
$K_l^{\text{glob}}, V_l^{\text{glob}}$
LM head
dec. layer $L/2$
dec. layer 1
$\times K$
$Z_0^{(1)} = H_E$
(d) Decoder layer, round t
softmax
H_E
$K^{\text{glob}}, V^{\text{glob}}$
S^{glob}
S^{loc}
P
output projection + residual
FFN (SwiGLU) + residual
$Z_l^{(t)}$
$Z_{l-1}^{(t)}$
$Q, K^{\text{loc}}, V^{\text{loc}}$
V^{glob}
V^{loc}